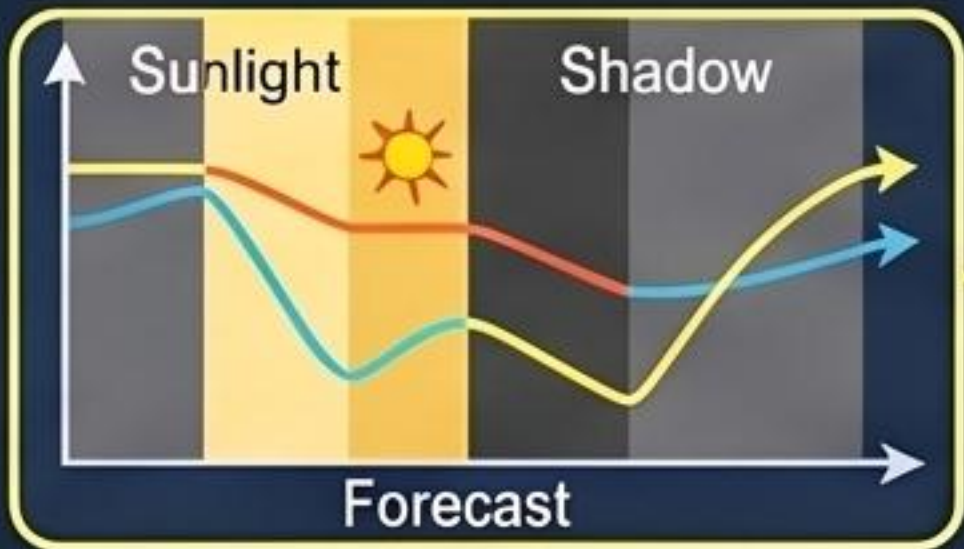


P-ECGR: Solving Energy Exhaustion in Deep Space SmallSat Networks

P-ECGR prevents relay battery failure and maximizes data delivery by forecasting future energy and using reservation-based booking, unlike resource-blind standard routing.

THE SOLUTION: PREDICTIVE & BOOKED ROUTING (P-ECGR)

Future Orbital Energy Prediction
Integrates solar-charging and eclipse windows to forecast battery levels at transmission time.



Priority-Based Load Balancing
Shifts low-priority traffic to high-capacity orbiters during a SmallSat's low-energy phases.

The "Energy Overbooking" Bottleneck
Standard routing ignores cumulative energy costs, causing SmallSat battery shutdowns and data loss.

The Conservatism Trap
Existing energy-aware systems avoid charging satellites, reducing overall network delivery efficiency.

PERFORMANCE PROOF: P-ECGR RESULTS

90.0% Delivery Ratio

P-ECGR outperforms standard and state-of-the-art algorithms in total data throughput.



Zero Packet Drops

Maintains perfect reliability while keeping SmallSat batteries above the 20% critical threshold.



P-ECGR PERFORMANCE VS. BASELINE

Metric	Standard CGR	SOTA ECGR	P-ECGR
Delivery Ratio	83.7%	89.5%	90.0%
Dropped Bundles	12.9	0.0	0.0
Avg. Latency	5483 s	5373 s	5073 s

Reservation-Based Energy Booking
A local registry "books" energy for committed bundles to prevent relay over-subscription.



12.9 Average Bundle Drops

Resource-blind routing frequently exhausts SmallSat batteries, leading to significant packet loss.